

AI-Powered Road Hazard Detection and Predictive Risk Scoring using YOLOv8

Abstract

RoadWatch AI is an intelligent road safety monitoring platform that combines YOLOv8-based computer vision, predictive analytics, and geospatial monitoring to detect potholes, cracks, sinkholes, and accident-prone zones in real time.

1. Introduction

Traditional road inspection methods are slow and inefficient. RoadWatch AI uses AI and deep learning to automatically monitor road damage, improve municipal response, and support safer urban infrastructure.

2. System Architecture

Data Acquisition: CCTV feeds, drone imagery, satellite images, and citizen reports.

Hazard Detection Engine: Detects potholes, cracks, and structural damage using YOLOv8.

Predictive Risk Scoring: Evaluates accident probability using traffic and weather data.

Dispatch System: Sends automated alerts to maintenance authorities.

3. Dataset & Training

The system was trained on 29,000+ annotated road images across multiple weather and traffic conditions. Data augmentation methods such as contrast enhancement, blur simulation, and rotation were applied.

4. YOLOv8 Enhancement

Enhanced YOLOv8n-seg architecture integrated DSConv, SimAM attention, and GELU activation functions for better boundary segmentation and small object detection.

5. Experimental Results

The proposed model achieved high detection accuracy with real-time performance suitable for smart city deployment.

6. Applications

RoadWatch AI supports smart city infrastructure management, accident prevention systems, automated hazard reporting, and municipal maintenance optimization.

7. Conclusion

The research demonstrates how AI-powered road monitoring systems can improve public safety, reduce infrastructure maintenance delays, and support data-driven governance.

Performance Metrics

Metric	Result
Best mAP@50	93.8%
Peak Precision	98%
Recall	93.3%
Real-Time FPS	85 FPS
Risk Prediction Accuracy	88.7%

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